

Python Project

Title: Student Management System (CSV Integration)

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Aim

To design and develop a beginner-friendly Student Management System in Python that demonstrates file handling with CSV, decision making, looping, arrays (lists), strings, and functions.

Procedure

Step 1: Create StudentManager Class

- The StudentManager class encapsulates all operations related to student records.
- It uses a CSV file (students.csv) to store and retrieve data.

Step 2: Add Student

- The add_student(name, roll, grade) method writes student details into the CSV file.
- The csv.writer object ensures proper formatting.
- A success message is displayed after adding a student.

Step 3: View Students

- The view_students() method reads all records from the CSV file using csv.reader.
- A loop prints each student's details.
- A try/except block handles missing file cases.

Step 4: Search Student

- The search_student(roll) method searches for a student by roll number.
- If found, details are displayed; otherwise, a "not found" message is shown.

Step 5: Generate Report

- The generate_report() method collects all grades into a list.
 - It prints the total number of students, the list of grades, and the highest grade.
 - Decision making ensures proper handling when no data exists.
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Code

```
import csv

class StudentManager:

    def __init__(self, filename="students.csv"):
        self.filename = filename

    def add_student(self, name, roll, grade):
        with open(self.filename, "a", newline="") as file:
            writer = csv.writer(file)
            writer.writerow([name, roll, grade])
            print(f"Student {name} added successfully!")

    def view_students(self):
        try:
            with open(self.filename, "r") as file:
                reader = csv.reader(file)
                print("\n--- Student Records ---")
                for row in reader:
                    print(f"Name: {row[0]}, Roll: {row[1]}, Grade: {row[2]}")
        except FileNotFoundError:
            print("No records found.")

    def search_student(self, roll):
        try:
            with open(self.filename, "r") as file:
                reader = csv.reader(file)
                for row in reader:
```

```
        if row[1] == roll:

            print(f"Found: Name={row[0]}, Roll={row[1]}, Grade={row[2]}")

            return

        print("Student not found.")

    except FileNotFoundError:

        print("No records found.")


def generate_report(self):

    try:

        grades = []

        with open(self.filename, "r") as file:

            reader = csv.reader(file)

            for row in reader:

                grades.append(row[2])

        if grades:

            print("\n--- Report ---")

            print(f"Total Students: {len(grades)}")

            print(f"Grades List: {grades}")

            print(f"Highest Grade: {max(grades)}")

        else:

            print("No data to generate report.")

    except FileNotFoundError:

        print("No records found.")
```

Output

```
Student Jai added successfully!
Student Surya added successfully!
Student Ayrus added successfully!

--- Student Records ---
Name: Jai, Roll: 101, Grade: A
Name: Surya, Roll: 102, Grade: B
Name: Ayrus, Roll: 103, Grade: A+

Found: Name=Surya, Roll=102, Grade=B

--- Report ---
Total Students: 3
Grades List: ['A', 'B', 'A+']
Highest Grade: B
```

Result

The Student Management System was successfully developed in Python.
It demonstrates:

- **File handling with CSV** for storing and retrieving student records
 - **Decision making** for search and report generation
 - **Looping** for displaying multiple records
 - **Arrays (lists)** for grade collection
 - **Strings** for handling names and roll numbers
 - **Functions and classes** for modular design
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